

Auteur: Yaman Haj Ahmad
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$$f(x) = \frac{1 + \tan(x)}{1 - \tan(x)}$$

$$\begin{aligned} f'(x) &= \frac{(1 - \tan(x)) (\sec^2(x)) - (1 + \tan(x)) (-\sec^2(x))}{(1 - \tan(x))^2} \\ &= \frac{(1 - \tan(x)) (1 + \tan^2(x)) + (1 + \tan(x)) (1 + \tan^2(x))}{(1 + \tan(x))^2} \\ &= \frac{(1 + \tan^2(x)) ((1 - \tan(x)) + (1 + \tan(x)))}{(1 + \tan(x))^2} \\ &= \frac{2(1 + \tan^2(x))}{(1 + \tan(x))^2} \end{aligned}$$

References